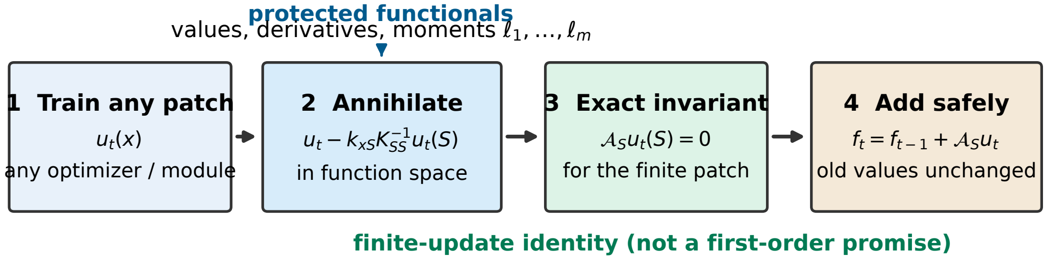
protected functionals
values, derivatives, moments $\ell_1, \ldots, \ell_m$
1 Train any patch
$u_t(x)$
any optimizer / module
2 Annihilate
$u_t - k_{xS}K_{SS}^{-1}u_t(S)$
in function space
3 Exact invariant
$\mathcal{A}_S u_t(S) = 0$
for the finite patch
4 Add safely
$f_t = f_{t-1} + \mathcal{A}_S u_t$
old values unchanged
finite-update identity (not a first-order promise)